



End-to-End AI Compilation for RISC-V: A Multi-Level Optimization Approach

RISC-V Summit Europe 2026 - Bologna



0 RuyiAI Overview

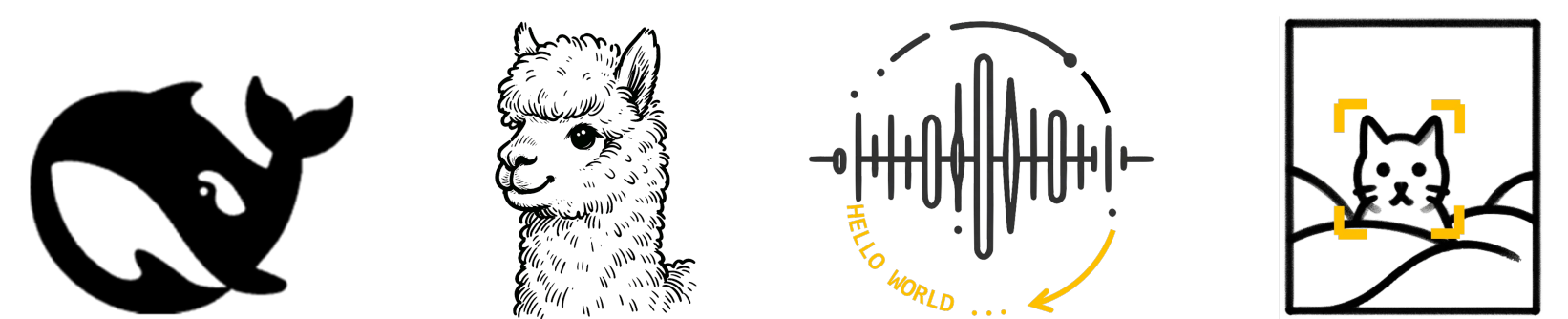
RuyiAI Homepage: <https://www.ruyiai.org/>

Buddy Compiler: <https://github.com/buddy-compiler/buddy-mlir>

RuyiAI is an AI system software stack designed to bridge the gap between AI models and RISC-V hardware platforms. It aims to establish a unified and extensible AI software ecosystem that addresses the fragmentation challenges in the current AI software stack. This poster focuses on the RuyiAI Buddy Compiler, the core compilation component of RuyiAI, which provides an end-to-end compilation pipeline from AI frameworks to RISC-V toolchains.

The RuyiAI Buddy Compiler interfaces with the PyTorch 2.x compilation framework and integrates DSL kernels via a graph infrastructure. The frontend imports PyTorch AI models and computation kernels into a unified graph representation. Moreover, the graph driver performs graph-level transformations and determines the compilation strategy, effectively converting the graph representation into MLIR.

1 RuyiAI Buddy Compiler



TorchDynamo Custom Compiler

Graph Capture → FX Graph → Aten IR

```
class GraphModule(torch.nn.Module):
    def forward(self, ...):
        ... = torch.ops.aten.arange.start(...)
        ... = torch.ops.aten.unsqueeze.default(...)
        ... = torch.ops.aten.view.default(...)
        ...
        return [... ...]
```

Graph Infrastructure

Graph Representation
(Ops Registry)

Fusion

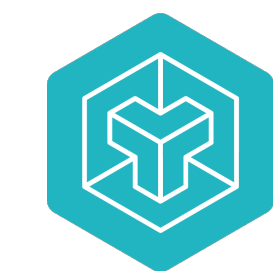
Replace

Graph Driver
(Compilation Strategies)

```
@triton.jit
def matmul_kernel(... ..):
    ... ..
    a = tl.load(... ..)
    b = tl.load(... ..)
    acc += tl.dot(a, b)
    ... ..
}
```

DSL Integration

High-Performance AI Kernels



Triton
↓
Triton TTIR



TileLang
↓
TVM TIR

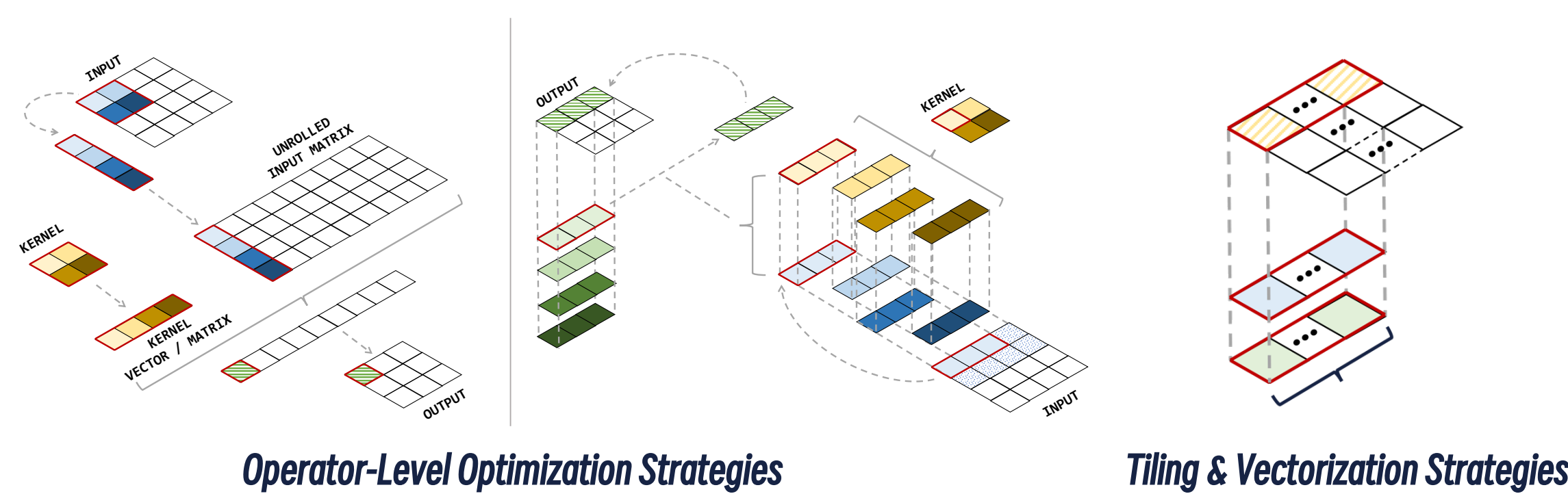


TReX
↓
RuyiAI TXIR

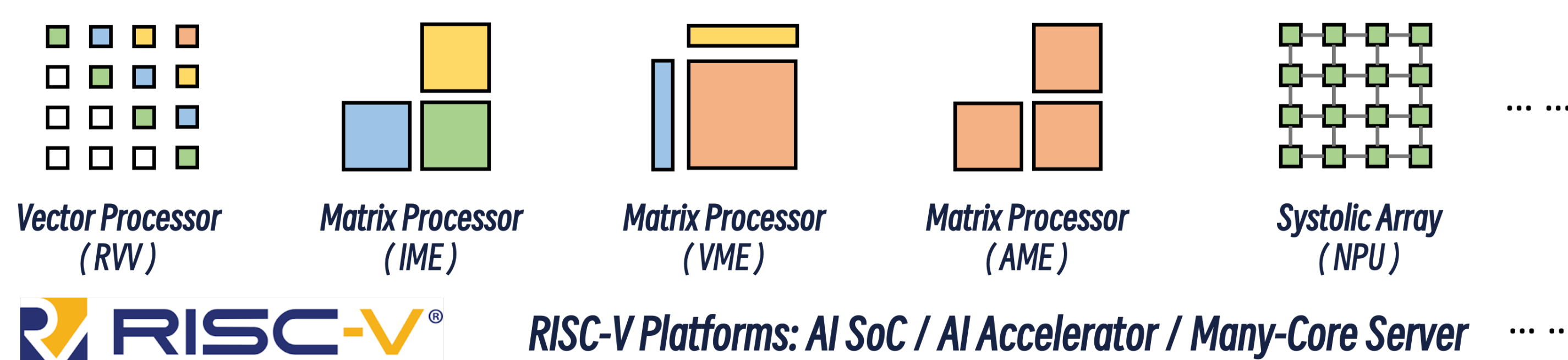
2 Multi-Level Compilation

The RuyiAI Buddy Compiler adopts a multi-level compilation strategy to progressively lower high-level operations into RISC-V specific operations. Multiple levels of optimization can be applied during this process, including tiling, operator-level optimizations, loop-level fusion, and vectorization. Most of these optimizations are closely related to hardware features. Therefore, the compilation passes are organized as configurable patterns that enable flexible transformation strategies.

MLIR High-Level Dialects & Custom Dialects



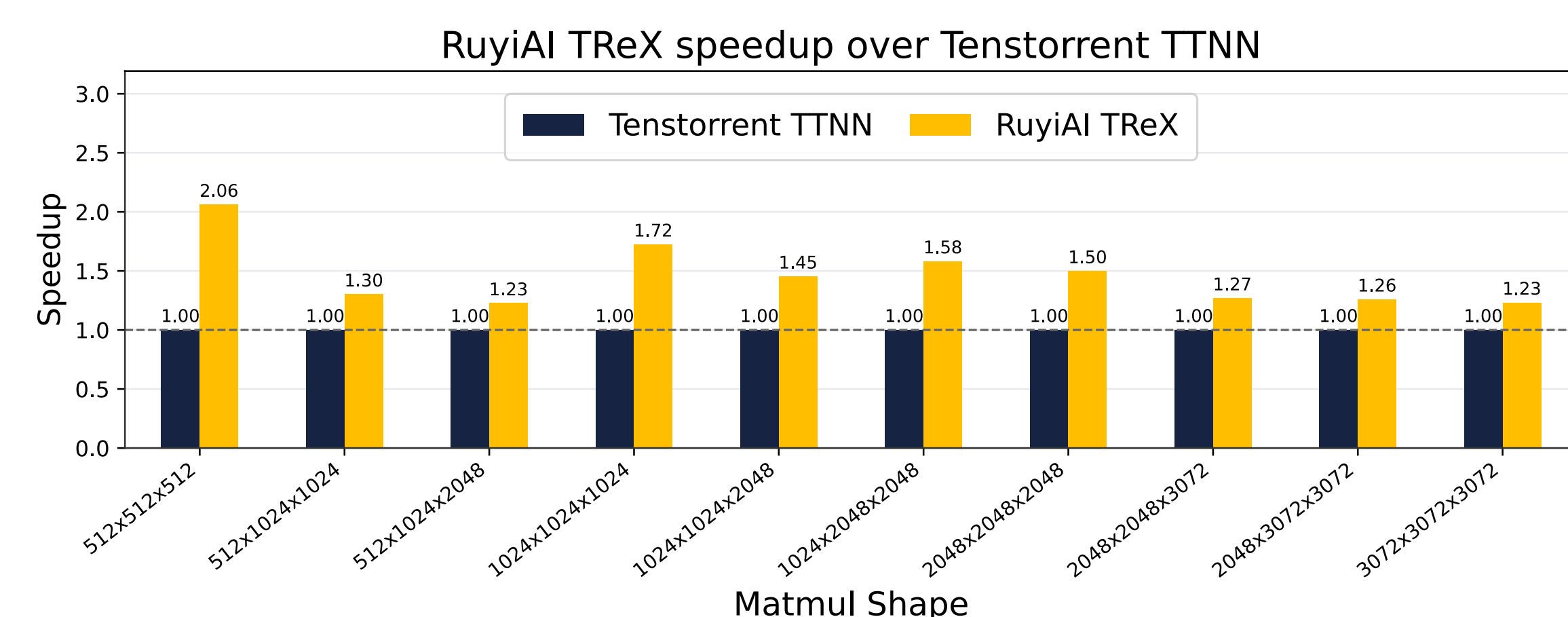
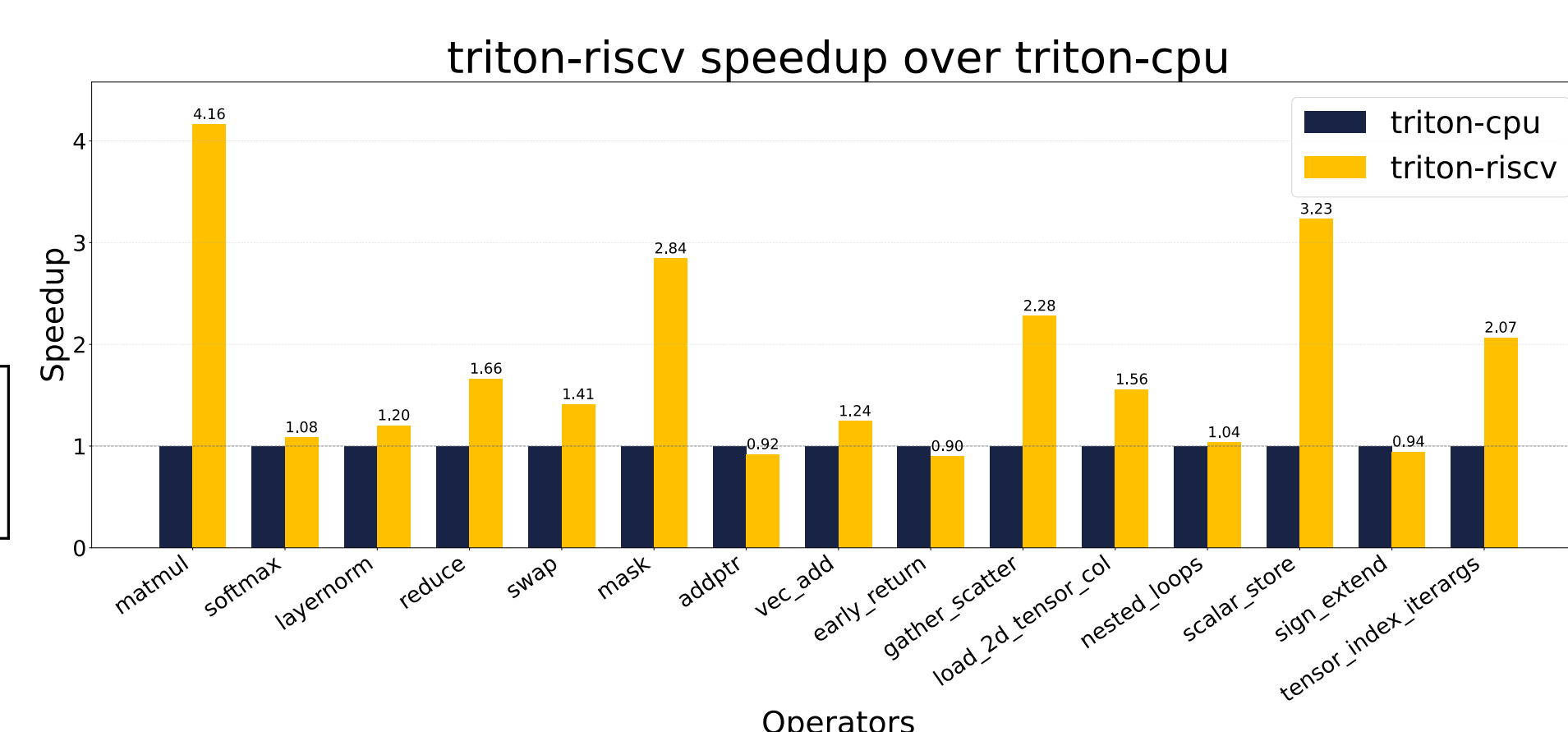
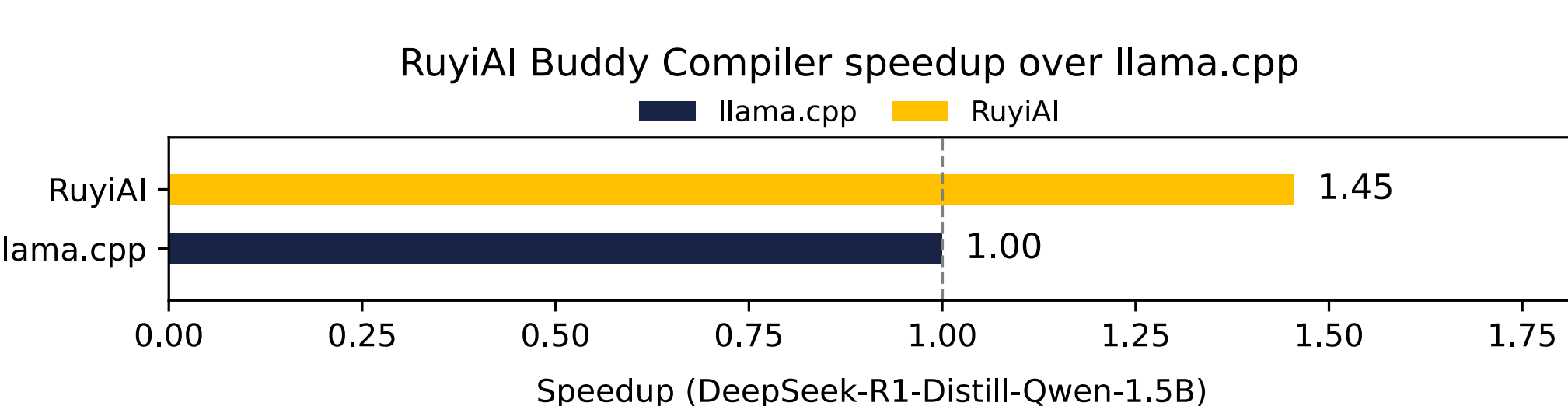
RVV / IME / VME / AME / Gemini Dialects



3 RISC-V Backend

The RuyiAI Buddy Compiler backends support multiple RISC-V extensions and accelerators, including RVV, IME, AME, and Gemini. These backends enable Buddy Compiler to generate efficient implementations for different RISC-V computing architectures, ranging from vector to matrix processors and accelerator-based designs. In addition, support for the upcoming VME extension will be incorporated once the specification is finalized.

4 Result & Evaluation



The RuyiAI Buddy Compiler supports approximately 85% of PyTorch operators, providing broad compatibility for model integration. On RISC-V CPUs, triton-riscv achieves significant speedups over triton-cpu through multi-level compilation optimizations. For end-to-end model inference, RuyiAI can reach a performance level comparable to llama.cpp and delivers better performance in specific cases. For RISC-V accelerators such as Tenstorrent, the RuyiAI Buddy Compiler and TReX Language have achieved better operator-level performance than the native software stack, while end-to-end inference optimization is being progressively integrated with multi-level compilation techniques.

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